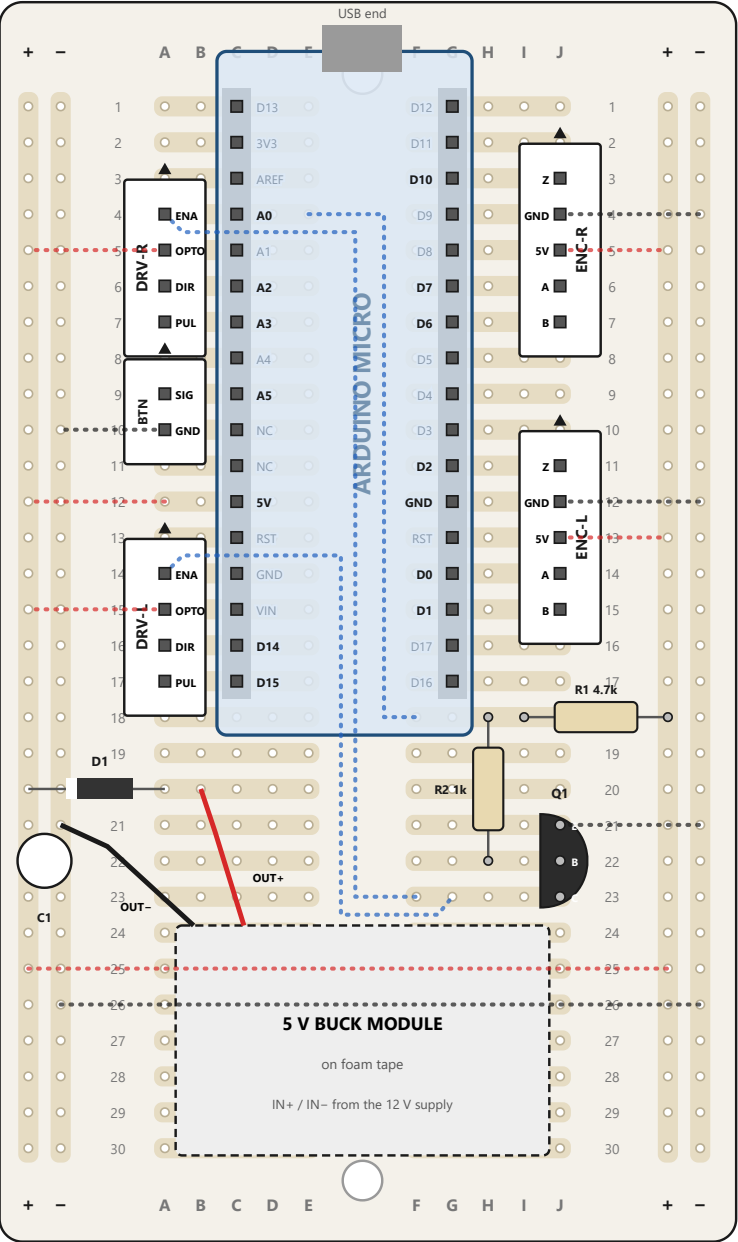




Placement (board coordinates as printed)

- Micro** 2 × 17-way female header sockets in **C1–C17** and **G1–G17**, USB end at row 1. Fit the top centre mounting screw first: it sits under the Micro.
- ENC-R** JST-XH 5-pin in **J3–J7**, pin 1 at J3. Right motor encoder.
- ENC-L** JST-XH 5-pin in **J11–J15**, pin 1 at J11. Left motor encoder.
- DRV-R** JST-XH 4-pin in **A4–A7**, pin 1 at A4. Right driver.
- DRV-L** JST-XH 4-pin in **A14–A17**, pin 1 at A14. Left driver.
- BTN** JST-XH 2-pin in **A9–A10**, pin 1 at A9. Start button (optional).
- D1** 1N5819: anode in **A20**, cathode (band) in the **left + rail** by row 20.
- C1** 100 µF 16 V across the left rails by row 22: + leg in the + rail.
- Q1** 2N2222 (or 2N3904): E in **J21**, B in **J22**, C in **J23**, flat face towards column I. Holds the drivers off until the program starts.
- R1 R2** R1 4.7 kΩ from **I18** to the **right + rail** by row 18. R2 1 kΩ from **H18** to **H22**.
- Buck** 5 V buck module on foam tape in the empty area (rows 24–30). OUT+ lead to **B20**, OUT– lead to the **left – rail** by row 21. IN+ / IN– from the 12 V supply (own pair).

Cables

Pin	ENC-R / ENC-L	DRV-R / DRV-L	BTN
1	YEL Z	ENA	button
2	BLK GND	OPTO	button
3	RED 5V	DIR	
4	BRN A	PUL	
5	BLU B		

- **The encoder pin order has changed** from the prototype: crimp to this table. Cut back ORG, GRN and WHT and insulate each.
- Driver cables follow the DM320T's P1 order. Motor leads go straight to the drivers.

How it works

- Each connector sits on the same strips as the Micro pins it uses, so **no signal needs a wire**: the copper joins them.
- Where a connector needs 5 V or GND instead, that strip is **cut** between the connector and the Micro, and a short link on the underside joins the connector pin to a rail (page 2).
- The Micro pins on cut strips (A1, VIN, the left GND, D9, D8, RST) are left unused. A0 is cut from DRV-R ENA and drives Q1 instead.

Connector	Pins reach (top to bottom)
ENC-R	D10, GND*, 5V*, D7, D6
ENC-L	D2, GND, 5V*, D0, D1
DRV-R	Q1 C (link 12)*, 5V*, A2, A3
DRV-L	Q1 C (link 11)*, 5V*, D14, D15
BTN	A5, GND*

* through a cut and an underside link

- Header or socket pin
- Link on the underside (page 2)

